DENTATRIAGE-NET

Automated Multi-Class Oral Pathology Detection System

Domain: Medical Computer Vision & Healthcare AI
Stack: Python, TensorFlow, Keras, EfficientNetB0, OpenCV, NumPy, Scikit-learn, Pandas

1. EXECUTIVE SUMMARY: SCALABLE MEDICAL INFERENCE

DentaTriage-Net is an end-to-end computer vision solution engineered for **High-Throughput Screening** and **Automated Triage** of oral pathologies. By leveraging State-of-the-Art (SOTA) Deep Learning architectures, the system classifies six clinically significant conditions: Caries, Calculus, Gingivitis, Ulcers, Tooth Discoloration, and Hypodontia.

The project addresses the "Data Velocity" problem in healthcare—where the volume of intraoral imagery far exceeds the human capacity for real-time review. Designed for population-scale health agencies, DentaTriage-Net identifies urgent cases from datasets of 10,000+ images, effectively eliminating diagnostic bottlenecks.

2. PROBLEM STATEMENT: CLINICAL DECISION FATIGUE

In large-scale screening programs, the primary challenge is the **Long-Tail Distribution** of disease and the resulting **Decision Fatigue**. When a clinician manually reviews 10,000 images, their sensitivity (Recall) naturally drops due to repetitive stress, leading to *False Negatives* in early-stage pathologies. DentaTriage-Net serves as a **Decision Support System (DSS)** that ensures a consistent, mathematically objective baseline across any dataset volume.

3. TECHNICAL ARCHITECTURE: EFFICIENTNETB0

Architectural Selection & Optimization

The model utilizes **EfficientNetB0**, selected for its superior **Floating Point Operations (FLOPs)** efficiency. Unlike standard CNNs, EfficientNet uses **Compound Scaling** to balance depth, width, and resolution. This ensures the model is lightweight enough for **Edge Deployment** on mobile dental cameras while maintaining high **Representational Power** for subtle oral textures.

Strategic Transfer Learning & Fine-Tuning

A two-phase strategy was implemented to leverage knowledge distillation:

- **Phase 1 (Feature Extraction):** The base model was frozen to preserve general spatial hierarchies while a custom Global Average Pooling and Dropout-regulated dense head were trained.
- **Phase 2 (Fine-Tuning):** The top 30 layers were unfrozen for specialized training using a **Differential Learning Rate (1e-5)**, adapting filters to high-frequency dental details.

4. DATA PIPELINE: ENGINEERING FOR RELIABILITY

The project involved a complex **Recursive Image Parsing** engine to handle unstructured directories. Key engineering implementations included:

- **Path-based Metadata Extraction:** Mapping inconsistent folder names to a unified Clinical Ontology.
- **Data Deduplication:** Implementing Hashed-Content Checks to remove redundant images across disorganized directories.
- **Stratified Splitting:** Ensuring the 70/15/15 split maintained identical class distributions to prevent bias in rare classes.
- **Advanced Augmentation:** Using Geometric (Rotation/Zoom) and Photometric (Brightness/Contrast) shifts to improve model generalization.

5. EVALUATION METRICS: THE RELIABILITY STANDARD

In medical imaging, accuracy is often misleading due to class imbalance. DentaTriage-Net is primarily evaluated using the **Weighted Macro F1-Score**.

While Precision ensures efficiency, **Recall (Sensitivity)** is prioritized to ensure no genuine pathology is missed. The F1-Score provides the harmonic mean, proving the model is a reliable tool for both common diseases (Caries) and rare conditions (Ulcers).

6. REAL-WORLD IMPACT & TRIAGE

For an agency managing 10,000 images, DentaTriage-Net provides:

- **Computational Triage:** Reducing the "Review Pool" from 10,000 to the ~1,000 most suspicious images.
- **Resource Allocation:** Generating statistical maps of disease prevalence across geographic sectors.
- **Patient Education:** Providing a "Visual Second Opinion" with Confidence Intervals to increase treatment compliance.

Computer Vision Engineer Skills (ATS Keywords)

Computer Vision: CNNs, EfficientNet, Transfer Learning, Feature Extraction, OpenCV, Image Augmentation.
Machine Learning: TensorFlow/Keras, Model Optimization, Hyperparameter Tuning, Stratified Sampling.
Evaluation: F1-Score (Macro/Weighted), Recall/Sensitivity, Precision, Confusion Matrix, ROC-AUC.